

Simon Lee

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EDUCATION

Columbia University, Columbia Business School

New York, New York

M.S. in Financial Economics

Aug 2026 – May 2028

- **PhD Coursework:** Advanced Derivatives, Econometrics and Statistical Inference, Computing for Finance

University of California, Los Angeles

Los Angeles, California

B.S. in Mathematics/Economics, Computing Specialization

Sep 2023 – Jun 2026

- Graduated in 3 years with 3.6 GPA while taking several PhD level courses, founding a club, and pursuing research
- **Coursework:** Graduate Machine Learning for Finance (A), Real Analysis (A), Graduate Cryptography (A)
- **Audited:** Graduate Probability Theory, Graduate Monte Carlo Methods for Optimization (P)

RESEARCH & EXPERIENCE

Bias Analysis for Langevin Monte Carlo

Los Angeles, California

Researcher

Apr 2026 – Present

- Sole-authored working paper giving the first two-sided total-variation bounds on the marginal discretization bias of the unadjusted Langevin algorithm for Gaussian targets; prior results (Chen et al., CPAM 2026; Lacker-Zhou 2025) are one-sided and in Wasserstein or entropy
- Computed the exact stationary law and closed-form marginal KL, sharpening the Gaussian-case rate from $O(hk)$ to $\Theta(h^2k)$, with $\Theta(\min(h\sqrt{k}, 1))$ in TV and an exact limiting TV profile in the isotropic case; no sparsity or banding assumed

Empirical Asset Pricing Research

Los Angeles, California

Researcher

Nov 2025 – Present

- Apply diffusion maps to 94 rank-normalized firm characteristics on CRSP monthly U.S. equities (1990-2018), compressing to 5 diffusion coordinates, with a PhD candidate and tenured faculty
- Engineered two embeddings: stock-level, yielding nonlinear betas with latent factors from ridge regression; and month-level, forecasting returns from softmax-weighted nearest-neighbor states over rolling 60-month windows
- Run the data pipeline and quantile-portfolio backtests against PCA and IPCA; vector diffusion maps and out-of-sample R^2 tests as next steps

Lagged Multi-Factor Model Research

Los Angeles, California

Researcher

Feb 2025 – Jan 2026

- Built a lead-lag detection pipeline for lagged multi-factor models with a UCLA mathematics postdoc, extending the subsequence clustering of Zhang et al. (2025)
- Estimated signed asset-factor exposures with neural networks under competing factor structures, clustered assets by DBSCAN on the induced cosine geometry, and recovered directed lead-lag links from lagged cross-correlations of cluster-level series; benchmarked against ONMF profiles on synthetic data

Citrix

Seattle, WA

Software Engineer Intern

Jun 2022 – Sep 2022

- Debugged and extended sample code for server load balancers with a distributed engineering team; presented implementations monthly to manager

INVOLVEMENT & LEADERSHIP

Bruin Algorithmic Traders

Los Angeles, California

Co-Founder, President and Director of Research

Feb 2025 – Jun 2026

- Practiced mathematical exposition giving weekly lectures on concepts in mathematical finance to junior club members
- Led 7 person research team focusing on engineering conditional factor models that use non-linear dimensionality reduction techniques; produced a paper draft accepted for publication in a UCLA-run Economics department journal

UCLA Phi Kappa Psi

Los Angeles, California

Social and Brotherhood Chair

Sep 2023 – Jun 2026

- Managed a \$30,000+ budget for various social, brotherhood and philanthropy related events

SKILLS & INTERESTS

- **Languages:** Python (primary); C++; LaTeX
- **Tools:** NumPy, pandas, scikit-learn, Git, CRSP/WRDS
- **Interests:** Cooking, Waterskiing, Mountaineering, Rubik's Cubes, Long Distance Running, Rock Climbing